A-A-50964
13 April 1984

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINE, TOOL ATTACHMENT, PRECISION

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification OO-G-673b.

This description covers electric motor driven tool attachment precision grinding machines intended for both external and internal grinding operations when attached directly to the compound slide of a lathe or when properly attached to other machine tools.

Salient characteristics.

Classification. The precision grinding machines covered by this CID shall be of the following types and of the sizes specified in Table I. The type and size machine required for a specific requirement shall be as specified in the contract or order.

Type I - A belt driven grinding machine which mounts onto the compound slide of a lathe or similar machine by means of a T-bolt mounting post.

Type II - A direct drive grinding machine which mounts into the tool post slot of a lathe or similar machine by means of a tool post shank.

Sizes and capacities. The Type I and Type II grinding machines shall meet the size and capacity requirements of Table I.

Design. The precision grinding machines covered by this CID shall be of the latest design, convenient to operate, and adaptable to a wide variety of grinding tasks when used with lathes, planers, milling machines, boring mills and similar machines.

Construction. The grinding machine shall be so constructed that when installed and connected to power, it shall be ready for operation. The grinding machine shall be capable of withstanding all forces encountered during grinding operations.

Materials. All parts of the grinding machine shall be constructed of the same materials and shall have the same physical properties, accuracies, and surface finishes as the parts used on the manufacturer's standard machine.

Electric motor and switch. The electric motor shall be the same as the electric motor supplied on the manufacturer's standard product provided that it meets the horsepower requirement of Table I. An electric switch of the proper electrical rating shall be provided to turn the grinding machine on or off.

FSC 3460

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TABLE I. Sizes and capacities

Type I Size	Type II Size	Grinding wheel dia. max. (inches) 1/		Horsepower	Spindle speed RPM (no load) 2/		Standard grinding wheels	
		External	Internal		Minimum	Maximum	Straight wheels Diameter and width (inches) 3/	Mounted wheels Diameter, width and length (inches)
1	--	2	2-1/4	1/14	10,000	22,500	2-1/4	1/4x1/4x1/8, 1/2x1/4x1/8, 2x1/4x1/8
2	--	3	1-1/4	1/4	6,000	40,000	1-1/4x1/4, 2x3/8, 3x3/8	1/4x1/4x1/8, 1/2x1/4x1/8, 3/4x1/4x1/8
3	--	4	1-1/4	1/3	5,000	45,000	1-1/2x1/4, 2x3/8, 3x3/8	1/4x1/4x1/8, 1/2x1/4x1/8, 1x1/4x1/4
4	--	5	1-1/2	1/2	4,200	45,000	1-1/2x1/4, 2-1/2x1/2, 4x1/2, 5x3/8	1/4x1/4x1/8, 1/2x1/4x1/8, 1x1/4x1/4
5	--	6	2-1/2	3/4	15,000	32,500	2x1, 2-1/2x3/8, 4x1/2, 5x1/2, 6x1/2	-----
6	--	8	5	1	2,600	28,000	3x3/4, 5x3/4, 8x3/4	-----
7	--	10	4-1/2	2	2,300	11,000	4x3/4, 4-1/2x3/8, 10x1-1/2	-----
8	--	12	5	3	2,000	11,000	2x1/2, 3x3/4, 5x3/4, 12x1/2	-----
9	--	14	6	5	1,700	11,000	2x1/2, 3-1/2x1/2, 5x3/4, 14x2	-----
--	1	6	1-1/2	1/4	3,000	4,000	1-1/2x3/8, 6x3/8	-----
--	2	8	2	1/2	3,000	4,000	2x1/2, 8x3/4	-----

1/ The maximum diameter of external and internal grinding wheels shall be not greater than the figures shown in the table for the type and size grinding wheel.

2/ The spindle speed range shall be from not more than the minimum shown to not less than the maximum shown for the type and size grinding machine required.

3/ The hole diameter of the standard wheels furnished shall be the same diameter as the spindle of the type and size grinding machine required.

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Electrical cord. Each grinding machine shall be provided with not less than six feet of flexible electrical cord, permanently attached to the machine with an appropriate electrical plug for connection to the power supply.

Electrical power. Grinding machines having 6 inch diameter grinding wheel and smaller shall operate on 115 volts AC, 60 Hz power. Grinding machines having 8 inch diameter grinding wheels and larger shall operate on 230 volts AC, 60 Hz, 3 phase power.

External and internal grinding. Each grinding machine shall be capable of switching from external to internal grinding operations. Each machine shall be furnished with the necessary arbors, spindle extensions, collet type chucks, or interchangeable spindles and quills to provide grinding with the size wheels listed in table I for the size grinding machine specified.

Drive pulleys and belts. The grinding machine shall be furnished with the manufacturer's standard set of drive pulleys and drive belts to provide all spindle speeds for which the grinder is designed.

Spindle. The spindle shall be accurately machined so that radial runout does not exceed .001 inch total indicator reading (TIR).

Bearings. The spindle shall be mounted in ball or tapered roller bearings. All bearings (except for sealed-for-life-types) shall be provided with accessible means of lubrication.

Belt tension. Means shall be provided to adjust belt tension.

Belt guard. Belt driven machines shall be equipped with a belt guard.

Threads. All threaded parts shall conform to FED-STD-H28.

Wheel guard. Each grinding machine shall be provided with a metal wheel guard. The guard furnished shall meet the design, dimensions, and wheel size requirements of ANSI B7.1.

Wheel collars. Metal flanges or collars shall be provided with the applicable spindle, spindle extension, or quill. The flanges or collars shall conform to the requirements of ANSI B7.1 and OSHA 2206. Means for securing the wheels and flanges to the work spindle shall be provided.

Vertical adjustment. Type I grinding machines shall be provided with a means to furnish vertical adjustment of the spindle for positioning to any lathe center.

T-bolt mounting post. Each type I grinding machine shall have the necessary T-bolt mount post for mounting the grinding machine onto the T-slot. T-bolt dimensions shall conform to ANSI B5.1.

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Tool post shank. Each type II grinding machine shall have the necessary tool post shank to securely attach the grinding machine to the machine tool post.

Grinding wheels. A grinding wheel of each size specified in table I for the size grinding machine required shall be furnished with each machine. The wheels furnished shall be general purpose, vitrified bond, medium grit, meeting the requirements of ANSI B7.1 and ANSI B74.13.

Diamond wheel dresser. A diamond wheel dresser shall be furnished with each grinding machine.

Wrench set. The manufacturer's standard set of wrenches shall be furnished with each grinding machine.

Tolerances. The grinding machine shall be capable of grinding work parts to size and roundness tolerances of 0.0005 inch and a surface finish of 10 micro-inches.

Carrying case. A metal carrying case shall be furnished with each grinding machine designed to drive grinding wheels 8 inches in diameter and smaller.

Electromagnetic interference suppression. When specified, machines procured under this CID shall comply with the electromagnetic interference control requirements of MIL-STD-461.

Safety devices. All machine parts and safety devices shall conform with the requirements of OSHA 2206 that are applicable to the machine itself.

Painting. The grinding machine shall be painted in accordance with indusdtrial standard practice.

Nameplate. A nameplate shall be securely attached to each grinding machine and shall contain name of manufacturer, model and serial number, minimum and maximum wheel size.

Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the CID where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to its being offered for acceptance under the contract. Quality conformance inspection shall consist of the following examination and test:

Examination. The grinding machine shall be examined to determined compliance with the requirements specified herein. Failure of the item to conform with any requirement of the CID shall be cause for rejection.

Test. Each grinding machine shall be operated at no load with a grinding wheel equal to the largest size which the machine will accommodate for not less than 15 minutes, including at least three starts. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period. Equipment requiring electromagnetic interference control testing shall be tested for compliance with MIL-STD-461.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes. American National Standard Institute ANSI B5.1 "T-Slots, Their Bolts, Nuts, Tongues, and Cutters, ANSI B7.1 "Safety Code for the Use, Care, and Protection of Abrasive Wheels", and ANSI B74.13 "Markings for Identifying Grinding Wheels and Other Bonded Abrasives" are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018. OSHA 2206 "General Industry, OSHA Safety and Health Standards (29 CFR 1910)" should be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

FED-STD-H28, Screw Thread Standards for Federal Services and MIL-STD-461 "Electromagnetic Interference Characteristics Requirements for Equipment" should be obtained from the contracting activity or as directed by the contracting officer.

Ordering Data. The contract or order shall specify:

(1) Quantity, type, and size grinding machine required.
(2) Distance from top of lathe tool compound slide to the lathe spindle and size of mounting post T-bolt required for Type I grinding machines. (These dimensions taken from lathe on which the grinding machine will be mounted.)
(3) Tool post slot size for type II grinding machines. (This dimension taken from machine on which grinding machine will be mounted.)
(4) Electromagnetic interference suppression, when required.

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Cross reference. The types and sizes listed in this CID are cross referenced to those listed in specification OO-G-673 as shown in Table II.

TABLE II. Cross reference.

OO-G-673 Type I		A-A-50964 Type I		OO-G-673 Type II		A-A-50964 Type II	
Size	Wheel dia. (inches)	Size	Wheel dia. (inches)	Size	Wheel dia. (inches)	Size	Wheel dia. (inches)
--	--	1	2	11	6	1	6
10	3	2	3	12	8	2	8
11	4	3	4	--	-	-	-
12	5	4	5	--	-	-	-
13	6	5	6	--	-	-	-
15	8	6	8	--	-	-	-
--	-	7	10	--	-	-	-
--	-	8	12	--	-	-	-
--	-	9	14	--	-	-	-

MILITARY INTEREST:

Custodians

Army - AL
Air Force - 99

Review Activities

DLA - GS, DH
Air Force - 84

User Activities

Navy - MC, SH

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT - NHT
GSA - FSS

PREPARING ACTIVITY:

DLA - IP

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